Requirement Elicitation: 'Bot vs Human'

Group 14

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Application purpose and the user group it serves.

The purpose of this application is to recognize bots from humans and filter them out. This application is for a group of people who want to have an analysis of their sessions and patterns that their clients used to buy tickets.

Functional requirements

The system shall allow users to provide an input-session in which all the variables are needed such as mouse behavior features, keyboard behavior features, activity patterns, and sessions meta data.
The system shall recognize and distinguish between humans and bots from the given input with 90% accuracy.
The system shall display the three most recent historically similar sessions including their labels compared to the input.
The system shall show a pattern for distinguishing among humans and bots.
The application shall allow users to query and filter the datasets using SQL syntaxes by having the users to select specific sessions and features related to bot activities .
The system shall display visualization of trends in the data through the graphs.

Key user questions

Is the user of the session a bot or a human?
What are the three most similar recorded sessions?
What data did the model use to show whether it is a bot or a human?

Machine learning goals

The goal for the machine is to be able to classify and flag suspicious user sessions that may be considered bots by recognizing similar and repeated patterns among the recorded sessions. This

will result in similar bot sessions being blocked and restricting their access to buy tickets, without causing any potential issue for real customers or having them flagged by mistake.

Dataset description

The dataset used by the system contains 400 recorded sessions of a ticket online web shop, collected by its anti-abuse team. The features mentioned in this dataset includes mouse behaviour, keyboard behaviour, activity patterns, and session meta data. The system shall use a column in this dataset as bot to display 1 or 0 if it is bot or human, respectively.